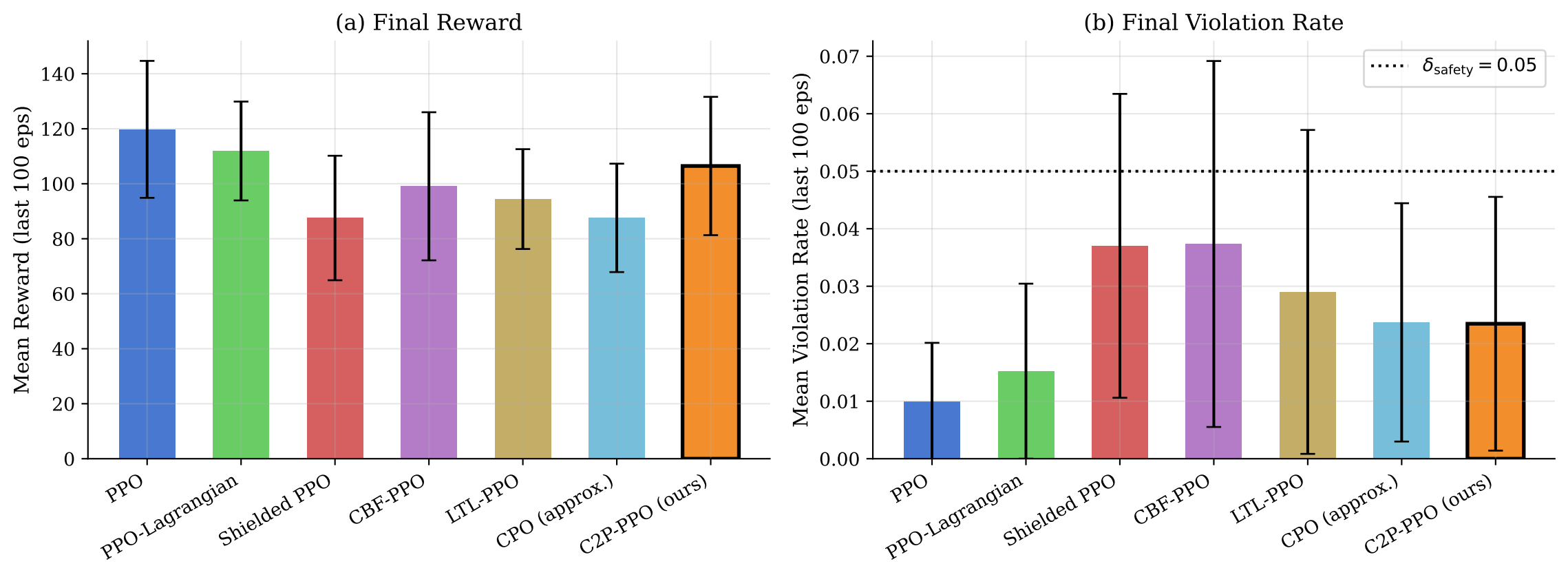
(a) Final Reward
Mean Reward (last 100 eps)
0
20
40
60
80
100
120
140
PPO
PPO-Lagrangian
Shielded PPO
CBF-PPO
LTL-PPO
CPO (approx.)
C2P-PPO (ours)
(b) Final Violation Rate
Mean Violation Rate (last 100 eps)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
$\delta_{\text{safety}} = 0.05$
PPO
PPO-Lagrangian
Shielded PPO
CBF-PPO
LTL-PPO
CPO (approx.)
C2P-PPO (ours)